TEST RESULTS

Table 4.1: Functional Evaluation of Clock Generation and Sequential Switching

Parameter	Method of Evaluation	Result Obtained
NE555 clock generation	Circuit powered and operation observed	Continuous clock operation was obtained
Tempo adjustment	Master tempo potentiometer adjusted during operation	Sequence rate increased or decreased with adjustment
CD4022 output switching	Q0–Q7 outputs observed during operation	Eight outputs activated sequentially
Step continuity	Complete sequence observed repeatedly	No skipped or repeated steps were observed
Clock-to-counter response	Counter observed while tempo control was varied	CD4022 remained responsive to the clock signal
Step transition	Sequential changes between outputs observed	Step transitions were clean and well defined

Table 4.2: Simulation Verification of the Main Circuit Functions

Function Simulated	Expected Behaviour	Simulation Result
NE555 clock generation	Continuous adjustable pulse output	Achieved
CD4022 sequencing	Sequential activation of eight outputs	Achieved
Step-to-step tone selection	Different resistance values selected for successive steps	Achieved
Tone generation	Distinct tone frequencies for different step settings	Achieved
Overall circuit operation	Clock, counter, and tone stages operate together	Achieved
Undesired behaviour	No major timing or noise-related anomaly	None of significance reported

Table 4.3: Comparison of Breadboard and Veroboard Implementation

Feature	Breadboard Prototype	Veroboard Implementation
Construction method	Temporary plug-in connections	Permanent soldered connections
Circuit operation	Functional	Functional
Eight-step sequencing	Achieved	Achieved
Tone generation	Audible and distinguishable	Retained after transfer
Pitch adjustment	Functional	Functional
Tempo adjustment	Functional	Functional

Continuity verification	Conducted during troubleshooting	Conducted before power-up
Mechanical stability	Temporary	Improved through permanent soldering

Table 4.4: Enclosure Implementation

Enclosure Requirement	**Implementation**	**Result**
Secure circuit housing	Veroboard fixed inside enclosure	Achieved
Tempo control access	Dedicated panel opening	Achieved
Eight pitch-control access points	Individual openings for pitch controls	Achieved
Power connection	External power-input opening	Achieved
Audio output	External audio-output opening	Achieved
Ventilation	Ventilated enclosure arrangement	Provided
Final wiring	Controls and connectors wired to circuit	Achieved
Post-assembly verification	Complete system tested after installation	System remained operational

Table 4.5: Final Prototype Performance Evaluation

Evaluation Parameter	**Method of Evaluation**	**Result**
Tempo range	Master tempo potentiometer adjusted during operation	Continuous and smooth tempo adjustment was obtained
Tuning response	Individual pitch trimmers adjusted for each step	Each of the eight steps responded independently to pitch adjustment
Step sequencing	Complete sequence observed repeatedly	All eight steps operated in the correct order
Step transitions	Audio output monitored during successive steps	Transitions were clean, with no audible glitching reported
Sound clarity	Audio output listened to during normal operation	Tones were clear, stable, and distinguishable
Audio output	Output connected to speaker/amplification equipment	Sufficient output was obtained for practical use
Usability	Normal operation of controls and interfaces	Controls operated as intended
Reliability	Repeated operation of completed prototype	Repeatable operation was obtained during testing